

ALTANFITH ALDWALIAH SPC

LIVE QUIZ PLATFORM

Technical & Commercial Proposal

Prepared for: **[Client Name]**

Reference: **[Proposal Reference]**

Version: **1.0**

Date: **[Date]**

CONFIDENTIAL

2. Confidentiality Notice

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3. Document Control

3.1 Document Information

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4. Table of Contents

- 2. Confidentiality Notice2
- 3. Document Control3
 - 3.1 Document Information3
 - 3.2 Revision History3
- 4. Table of Contents4
- 5. Letter of Submission6
- 6. Executive Summary7
- 7. About Altanfith Aldwaliah SPC9
- 8. Understanding of Client Requirements10
- 9. Business Challenges11
- 10. Proposed Solution12
- 11. Platform Overview13
- 12. Current Use Case14
- 13. Functional Scope15
 - 13.1 Quiz Creation & Content.....15
 - 13.2 Live Hosting & Participation16
 - 13.3 Branding & Experience16
 - 13.4 Reporting & Analytics.....17
 - 13.5 Administration & Governance18
 - 13.6 AI Capabilities19
- 14. AI Capabilities20
 - 14.1 Current Capabilities (Included)20
 - 14.2 Optional Capabilities (Scoped per Engagement).....20
 - 14.3 Future Roadmap.....21
- 15. User Roles22
- 16. Reporting and Analytics23
- 17. Technical Architecture25
 - 17.1 Technology Stack25
 - 17.2 Architecture Diagram.....25
- 18. Real-Time Architecture27
 - 18.1 Session Flow.....27
 - 18.2 Server-Authoritative Timing, Reconnection, and Why It Matters at Scale27

19. Deployment Options.....	28
19.1 Cloud / SaaS.....	28
19.2 On-Premises.....	28
19.3 Hybrid	29
20. Cybersecurity.....	30
21. Data Protection.....	31
22. Data Ownership	32
23. Integration Options.....	33
24. Performance and Scalability.....	34
25. Implementation Methodology	35
26. Project Timeline	37
26.1 Milestones	37
27. Project Governance	39
28. Deliverables	40
29. Training and Knowledge Transfer	42
30. Testing and Acceptance	43
31. Support and Maintenance.....	44
32. SLA.....	45

5. Letter of Submission

[Client Name]

[Date]

Dear Sir / Madam,

Subject: Proposal for the Supply, Configuration, and Support of the Live Quiz Platform

On behalf of Altanfith Aldwaliah SPC, I would like to thank **[Client Name]** for the opportunity to submit this proposal. We appreciate the time and consideration your team has invested in reviewing our solution, and we welcome the opportunity to support your organization's engagement, training, and assessment objectives.

This proposal presents the Live Quiz Platform as an enterprise engagement, assessment, and analytics solution designed to help organizations such as **[Client Name]** run live sessions, structured assessments, and knowledge-evaluation activities with the branding control, reporting depth, and security posture that a large organization requires — going beyond what general-purpose public quiz tools are built to offer. The following sections set out our understanding of your requirements, the proposed solution, our technical architecture and delivery methodology, and the commercial models available for this engagement.

This proposal, including its commercial terms, is valid for a period of 30 days from the date shown above. We remain available throughout this period to answer questions, arrange a live demonstration, or refine any aspect of this proposal to better reflect **[Client Name]**'s specific requirements.

We look forward to the possibility of working with **[Client Name]** and would welcome the opportunity to discuss this proposal further at your convenience.

Yours faithfully,

Haitham Al Lamki

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Sultanate of Oman

6. Executive Summary

Altanfith Aldwaliah SPC is pleased to present the Live Quiz Platform to **[Client Name]** as an enterprise engagement, assessment, and knowledge-evaluation platform — a flexible, customizable alternative to general public quiz tools, purpose-built for organizations that require organizational branding, structured reporting, and enterprise-grade controls around how live sessions and assessments are run.

The platform is not a concept or a prototype. It is already in real operational use at Abraj Energy Services, where it supports HSE meetings and employee engagement activities, with single live sessions that have brought together more than 100 participants at once. This production track record gives **[Client Name]** confidence that the core platform is stable, functional, and capable of supporting real organizational events at meaningful scale, rather than a solution that exists only on paper.

The platform is bilingual by design, with a fully localized Arabic and English interface including complete right-to-left (RTL) support for Arabic — a capability that matters for organizations operating across Oman and the wider Gulf region, where sessions, dashboards, and reports need to be equally usable in either language.

Content creation is accelerated through AI-assisted generation: administrators can generate a complete quiz from a topic or title, or from an uploaded PDF, Word document, URL, or pasted text, with the AI assigning suggested difficulty levels and answer explanations. Every AI-generated item is presented to an administrator for review and editing before it is used in a live session — the platform is designed to keep a human reviewer in the loop rather than publishing AI output automatically. Additional AI capabilities, such as AI-assisted rewriting, suggestions, and generated visual themes, are available as optional, separately scoped additions, and a further set of AI-driven insights and reporting features sits on our future roadmap.

[Client Name] can choose how the platform is hosted and delivered. Three deployment models are available: a fully managed cloud/SaaS option, a fully on-premises option in which the Express.js backend runs entirely within the client's own infrastructure, and a hybrid option that splits components between the two — the right choice depends on **[Client Name]**'s data residency, network, and cybersecurity requirements, and is confirmed through a dedicated infrastructure assessment rather than assumed in advance. Commercially, the platform is available under three models — an annual SaaS subscription, a one-time perpetual license, and an enterprise custom development partnership — so that **[Client Name]** can select the ownership and cost structure that best fits its procurement approach.

Underpinning all of this is the fact that Altanfith Aldwaliah SPC is an Omani company: delivery, support, and communication happen locally, in Arabic and English, without the delays or handover friction that come with an offshore vendor. This local presence supports the value of the platform — it is the foundation the offer is delivered on, not the offer itself.

Proposal at a glance	
Solution	Enterprise engagement, assessment & analytics platform — Live Quiz Platform
Proven use	In production at Abraj Energy Services — HSE meetings & engagement sessions with 100+ live participants
Deployment	Cloud/SaaS, fully on-premises, or hybrid — confirmed via infrastructure assessment

Commercial models	Annual SaaS subscription · perpetual license · enterprise custom development partnership
AI	AI-assisted quiz generation from topic, PDF, document, URL, or text — always reviewed by an administrator before use
Support	Bronze / Silver / Gold packages, delivered locally from Oman in Arabic and English

7. About Altanfith Aldwaliah SPC

Altanfith Aldwaliah SPC is a Sultanate of Oman-based software company focused on designing, building, and supporting enterprise digital platforms for government, oil & gas, education, and private-sector organizations. Our work centers on delivering software that is not simply installed and left running, but actively configured, branded, and adapted around each client's operational, reporting, and security requirements.

Our core capability areas span custom application development, enterprise platform configuration, and applied artificial intelligence — including AI-assisted content generation of the kind used within the Live Quiz Platform itself. We combine this technical capability with a practical understanding of how large Omani organizations operate: the governance structures they answer to, the security review processes their IT and cybersecurity teams require, and the bilingual Arabic/English environment in which their staff and stakeholders work.

The Live Quiz Platform is a direct expression of this approach. Rather than offering a generic, one-size-fits-all quiz tool, we designed and continue to develop a platform that can be shaped around each client's branding, workflows, integration landscape, and deployment preferences — and that is already proven in real operational use, as set out in Section 12 of this proposal.

As an Omani company, Altanfith Aldwaliah SPC is able to engage directly with **[Client Name]**'s teams throughout discovery, delivery, and ongoing support, in Arabic and English, without reliance on an intermediary or an offshore support desk operating outside Gulf business hours.

8. Understanding of Client Requirements

Large Omani organizations — whether in government, oil & gas, education, or the private sector — typically approach an engagement and assessment platform with a common set of expectations that go well beyond what a free or public quiz tool is designed to provide. Based on our experience delivering the Live Quiz Platform into a live production environment, we understand these expectations to include the following.

Requirement	Our response
Full control over branding, so that sessions and materials reflect the organization's own visual identity rather than a generic third-party look	Per-tenant organization branding, including logo and color customization, custom themes, custom backgrounds, and custom UI text, is built into the platform
Control over where data resides and how it is retained, including the ability to keep sensitive data within the organization's own environment	A fully on-premises deployment option is available, in which the backend and database can run entirely within [Client Infrastructure] ; data retention and export terms are defined in the commercial agreement, and final data residency arrangements are confirmed during discovery and infrastructure assessment
A genuinely bilingual Arabic and English experience, not a partial translation layered onto an English-first product	The platform interface, including full right-to-left (RTL) layout, is available in Arabic and English as a core, included capability
Compatibility with the organization's internal cybersecurity review and approval process before go-live	The platform is developed with awareness of the OWASP Top 10 and includes controls such as role-based access, audit logging, and API rate limiting; the full architecture is made available for client security review, and deployment is subject to the client's cybersecurity assessment as described in Section 20
Practical integration paths into the organization's existing systems (identity, HR, reporting) rather than a fully closed platform	REST APIs and a defined set of integration options are available; specific integrations such as single sign-on, HR systems, or Power BI can be configured to support [Client Name] 's environment and are scoped individually, as described in Section 23
Local support that understands the organization's language, working hours, and escalation expectations	Altanfith Aldwaliah SPC is based in the Sultanate of Oman and delivers support in Arabic and English under Bronze, Silver, and Gold packages, as described in Section 31

These requirements shape both the design of the platform and the structure of this proposal: the sections that follow set out how the Live Quiz Platform, its architecture, and our delivery methodology are built to answer each of them directly.

9. Business Challenges

Organizations that rely on free or general-purpose public quiz tools for internal engagement, training, or assessment activities commonly encounter a consistent set of limitations once usage moves beyond a single informal session.

Challenge	Consequence
Generic public quiz tools are built for open, consumer use and lack enterprise controls	No organizational branding, no role-based administration, and no ability to enforce the organization's own access and data-handling requirements
Quiz and question content is created ad hoc and scattered across individual accounts, files, and departments	Content cannot be reused, standardized, or governed centrally, and is often lost when the individual who created it moves on
Public tools offer little to no organizational-level reporting	Management cannot see participation, performance, or engagement trends across departments, sessions, or time — results exist only as isolated, per-session snapshots
Data created and collected through a public, third-party tool sits outside the organization's own environment and control	This raises data residency and governance concerns for organizations subject to internal compliance requirements or operating in regulated sectors, particularly where session content or participant data is sensitive
Repetitive use of the same generic tool and format across HSE briefings and training sessions leads to engagement fatigue	Participation and attention decline over time, undermining the safety and learning objectives the sessions are intended to support
Public tools typically offer only a single, English-first interface	Arabic-speaking participants and administrators are left with a translated or inconsistent experience, reducing usability and adoption across a bilingual workforce

The Live Quiz Platform, and the enterprise delivery model set out in this proposal, are designed specifically to address these gaps — as described in the sections that follow.

PART TWO

How the Live Quiz Platform answers the requirements and challenges set out above

10. Proposed Solution

Altanfith Aldwaliah SPC proposes the Live Quiz Platform as a single platform that brings together live engagement, structured assessment, and organizational analytics, so that **[Client Name]** does not need to operate separate tools for running a live session, evaluating knowledge, and reporting on the results.

The platform is designed to support a broad range of use cases across a large organization, from safety and training activities to recruitment and internal communication. The table below reflects the categories of use the platform is built to support; the specific mix relevant to **[Client Name]** can be confirmed during discovery.

Engagement & Training	Assessment & Evaluation	Organizational & Institutional
HSE meetings	Knowledge assessments	Government awareness campaigns
Employee training	Recruitment assessments	Educational institutions
Corporate workshops	Interviews	Schools & universities
Team-building	Surveys & polls	Internal compliance assessments
Corporate events	Certification preparation	Employee onboarding
Competitions	Learning & development programs	Safety awareness

It is important to be clear about how we position the Live Quiz Platform. Live, buzzer-style quiz tools such as Kahoot are widely known and are a useful category reference for the live-session experience — but the Live Quiz Platform is not a copy of any such tool. It is an enterprise platform: customizable to **[Client Name]**'s operational workflows, branding, reporting requirements, security posture, and integration landscape, and deployable in a way a general public tool is not, including fully on-premises within the client's own infrastructure. The sections that follow set out the platform itself (Sections 11–16), its technical architecture and deployment options (Sections 17–24), our delivery methodology (Sections 25–32), and the commercial terms under which it is offered (Sections 33 onward).

11. Platform Overview

The Live Quiz Platform is best understood through the lifecycle of a typical session, from creation through to reporting.

An administrator or quiz creator begins by building a quiz manually, or by using the platform's AI-assisted generation to produce a complete quiz from a topic, an uploaded PDF or Word document, a URL, or pasted text — with every AI-generated question reviewed and editable before use. The quiz is then branded with the organization's own logo, colors, and theme, and can be organized into a reusable question bank tagged by subject and difficulty for future sessions.

When it is time to run a session, the host launches the quiz live. Participants join in seconds using a QR code, a direct link, or a session PIN, without needing to create an account. As the session runs, the platform is server-authoritative for timing and scoring: the server determines when a question opens and closes and calculates results, while participant and host devices render what is happening in real time — including live answer submission, a live leaderboard, and automatic reconnection if a participant's network connection drops mid-session. At the close of the session, the top three participants are shown on a podium, alongside the full participant ranking.

After the session ends, results are not lost. They are captured in the platform's reporting and session history, giving administrators and management access to participation data, performance analysis, and historical comparisons across sessions — described in full in Section 16.

Across this lifecycle, the platform's capability set is organized into six pillars, summarized below and detailed in full in Section 13.

Pillar	What it covers
Quiz creation & content	Quiz and question authoring, question banks, bulk import, templates, versioning, and auto-save/draft recovery
Live hosting & participation	QR/link/PIN joining, real-time answers, live leaderboard, ranking, podium, and automatic reconnection
Branding & experience	Organization branding, custom themes and backgrounds, and a bilingual Arabic/English interface with full RTL support
Reporting & analytics	Result summaries, performance and difficulty analysis, historical and department-level reporting, and Excel/CSV/PDF export
Administration & governance	Role-based access, multi-tenant organization/department structure, multiple administrators, audit logs, and session history
AI capabilities	AI-assisted quiz generation from topic, document, URL, or text, with administrator review before use — detailed in Section 14

12. Current Use Case

The Live Quiz Platform is not a concept, prototype, or proof of demonstration — it is in real, ongoing operational use today.

The platform is currently deployed and actively used within **Abraj Energy Services**, where it supports HSE (Health, Safety, and Environment) meetings and employee engagement activities. Live sessions run on the platform have brought together more than 100 participants at once, joining and answering in real time within a single session.

We present this use case as evidence that the core platform — live session hosting, real-time participation at meaningful scale, and the underlying engagement experience — is production-proven and functioning in a real organizational setting, rather than as a marketing claim or a controlled demonstration environment. We have not included any additional client references, dates, or testimonial statements beyond what is described above, and we will not represent any other deployment as a live production reference unless and until it is one.

13. Functional Scope

The tables below set out the platform's functional scope in full, grouped into six capability areas. Each capability is marked with its current status: **Included** capabilities are part of the core platform today; **Optional** capabilities are available as scoped additions; **Future enhancement** items are on our roadmap and not part of the current platform.

13.1 Quiz Creation & Content

Capability	Description	Status
Quiz creation & management	Create, edit, and manage quizzes from a central administration interface	Included
Multiple question types	Single-answer questions with 4–6 choices, multi-select questions, and true/false questions	Included
Polls & surveys	Non-scored question formats for gathering opinion or feedback	Included
Question timers	Configurable time limits per question	Included
Scoring & points configuration	Configurable points and scoring rules per question or quiz	Included
Public & private quizzes	Quizzes can be restricted to the organization or made available more broadly, per administrator configuration	Included
Question bank	Central, reusable question repository organized by subject, tags, and difficulty	Included
Bulk import	Bulk import of questions from Excel, CSV, or Word files	Included
Quiz templates & reuse	Reusable quiz templates to standardize content across sessions and departments	Included
Edit & duplicate quizzes	Existing quizzes can be edited or duplicated as a starting point for new sessions	Included
Auto-save during creation	Quiz content is saved automatically while an administrator is authoring it	Included

Draft recovery	In-progress drafts can be recovered rather than lost	Included
Versioning of quiz content	Changes to quiz content are versioned over time	Included

13.2 Live Hosting & Participation

Capability	Description	Status
Live quiz hosting	Host a quiz as a live, timed session for a group of participants	Included
QR-code joining	Participants join a session by scanning a QR code	Included
Direct-link joining	Participants join via a direct shareable link	Included
Game code / session PIN	Participants join by entering a short session PIN	Included
Real-time participant joining	Participants appear in the host's lobby as they join, in real time	Included
Real-time answer submission	Participant answers are submitted and processed in real time via WebSocket connection	Included
Real-time leaderboard	A live leaderboard updates as answers are scored	Included
Live ranking	Participant ranking updates live throughout the session	Included
Top-three winners podium	A podium view highlights the top three participants at session close	Included
Full participant ranking	Complete ranking of all participants, not only the top three	Included
Automatic reconnection	Participants who lose network connectivity mid-session are automatically reconnected	Included

13.3 Branding & Experience

Capability	Description	Status
Custom themes	Visual themes can be selected or configured per organization	Included
Custom backgrounds	Custom background imagery for quizzes and sessions	Included
Organization branding	Per-tenant branding, including logo and color customization	Included
Custom UI text	Interface text can be adapted per organization	Included
Arabic & English interface with full RTL	Complete bilingual interface with full right-to-left layout support for Arabic — positioned as a core enterprise bilingual capability, not an add-on translation layer	Included
Additional languages	Support for interface languages beyond Arabic and English	Optional

13.4 Reporting & Analytics

Capability	Description	Status
Quiz result summary	Summary results at quiz and session level	Included
Participant ranking	Ranked results per participant	Included
Top performers	Identification of top-performing participants	Included
Correct/incorrect answer analysis	Breakdown of correct versus incorrect responses	Included
Question difficulty analysis	Analysis of question-level difficulty based on response patterns	Included
Participant completion	Visibility into who completed a session and who did not	Included
Department-level reporting	Reporting rolled up by department or organizational unit	Included

Historical reports	Access to results and reports from past sessions	Included
Session comparison	Comparison of results across multiple sessions	Included
Export to PDF	Report export in PDF format	Included
Export to Excel & CSV	Report export in Excel and CSV formats, with Arabic and English report headers	Included
Management dashboard	Consolidated dashboard view for management-level oversight	Included
AI-generated insights	AI-generated narrative insights over historical results	Future enhancement

13.5 Administration & Governance

Capability	Description	Status
Administrative dashboard	Central dashboard for platform and content administration	Included
Organization user accounts	User accounts scoped to the organization	Included
Role-based access	Access and permissions governed by role-based access control	Included
Multiple departments/organizations	Multi-tenant support for multiple departments or organizations on one platform, each with isolated data	Included
Multiple administrators	Support for more than one administrator per organization	Included
Audit logs of administrative actions	Administrative actions are recorded in an audit log	Included
User activity/login visibility	Visibility into user activity and login events	Included
Quiz session history	Historical record of quiz sessions run	Included

Soft-delete with archive	Deleted content is archived rather than permanently removed, supporting recovery	Included
Notification options	Configurable notifications for administrators and/or participants	Optional

13.6 AI Capabilities

Capability	Description	Status
AI-assisted content generation	Generate quizzes from a topic, an uploaded PDF or Word document, a URL, or pasted text, with administrator review before use. The full classification of current, optional, and future AI capabilities is set out in Section 14 (AI Capabilities)	See Section 14

14. AI Capabilities

The Live Quiz Platform includes AI-assisted content generation as a core, already-implemented capability, alongside a further set of AI features that are available as optional, separately scoped additions, and a set of AI-driven reporting capabilities that sit on our future roadmap. This section sets out the distinction clearly, so that **[Client Name]** can see exactly what is included today, what can be added, and what is planned for the future.

14.1 Current Capabilities (Included)

Capability	Description	Status
Quiz generation from a topic or title	A complete quiz, including questions and answer choices, is generated from a short topic or title prompt	Included
Quiz generation from an uploaded PDF	Questions are generated directly from the content of an uploaded PDF document	Included
Quiz generation from an uploaded document	Questions are generated directly from the content of an uploaded Word document	Included
Quiz generation from a URL	Questions are generated from the content of a supplied URL or webpage	Included
Quiz generation from pasted text	Questions are generated directly from text pasted into the platform	Included
AI-assigned difficulty levels	Generated questions are automatically assigned a suggested difficulty level	Included
AI-generated answer explanations	Generated questions include a suggested explanation for the correct answer	Included
Administrator review before use	All AI-generated content is presented to an administrator for review and editing before it is used in a live session or published to a question bank	Included

14.2 Optional Capabilities (Scoped per Engagement)

The following AI capabilities are not part of the current platform but can be added as a separately scoped engagement.

Capability	Description	Status
AI question suggestions	AI-suggested questions offered while an administrator authors a quiz manually	Optional
AI answer suggestions	AI-suggested answer choices for a manually authored question	Optional
AI question rewriting	AI-assisted rewriting of existing question text for clarity or tone	Optional
AI-assisted difficulty re-adjustment	AI-assisted review and re-adjustment of difficulty levels across an existing question bank	Optional
AI-generated quiz backgrounds	AI-generated background imagery for quizzes and sessions	Optional
AI-generated themes	AI-generated visual themes for the platform interface	Optional
AI-assisted result summaries	AI-assisted narrative summary of a single session's results	Optional

14.3 Future Roadmap

Capability	Description	Status
AI-assisted insights over historical results	AI-generated narrative insights and trend analysis across historical session data	Future enhancement
AI-generated management reports	AI-generated summary reports intended for management-level review	Future enhancement

AI features are delivered using a commercial large-language-model API; usage allowances, and any associated API costs, are defined in the applicable commercial model (see Section 33). Cloud-based AI generation requires internet connectivity. A private, on-premises AI deployment is a future roadmap option that would depend on separate, dedicated infrastructure and is not part of the current platform. As with any AI-generated content, **[Client Name]** should review AI-generated quizzes, questions, and summaries before they are used in a live or published context.

15. User Roles

The platform's role-based access model is designed to reflect how a large organization actually assigns administrative and operational responsibility, rather than offering a single flat administrator account.

Role	Purpose
Super Administrator	Full platform administration across all organizations and departments
Organization Administrator	Manages a single organization, including its users, branding, and content
Department Administrator	Manages a department's content and users within an organization
Quiz Creator	Authors quizzes and question banks
Trainer	Runs training-oriented sessions
Host	Runs live sessions
Viewer	Views results
Participant	Joins and answers in a live session — no account required
Auditor	Read-only access to audit and activity logs
Read-Only User	Read-only access to dashboards and reports

The final role model applied for **[Client Name]** is configured during discovery, and can be mapped to the client's own directory groups where a single sign-on integration is included in scope (see Section 23).

16. Reporting and Analytics

The platform's reporting and analytics capabilities give administrators and management visibility into participation, performance, and engagement, both at the level of a single session and across the organization's session history over time.

Capability	Description	Status
Quiz result summary	Summary results at quiz and session level	Included
Participant ranking	Ranked results per participant	Included
Top performers	Identification of top-performing participants	Included
Correct/incorrect answer analysis	Breakdown of correct versus incorrect responses	Included
Question difficulty analysis	Analysis of question-level difficulty based on response patterns	Included
Participant completion	Visibility into who completed a session and who did not	Included
Department-level reporting	Reporting rolled up by department or organizational unit	Included
Historical reports	Access to results and reports from past sessions	Included
Session comparison	Comparison of results across multiple sessions	Included
Export to PDF	Report export in PDF format	Included
Export to Excel & CSV	Report export in Excel and CSV formats	Included
Management dashboard	Consolidated dashboard view for management-level oversight	Included
AI-generated insights	AI-generated narrative insights over historical results	Future enhancement

All PDF, Excel, and CSV exports are compliance-ready: report headers are rendered in both Arabic and English, so that reports generated for **[Client Name]** can be shared and archived internally without additional reformatting, regardless of which language the reviewing team works in.

PART THREE

How the platform is built, deployed, and secured

17. Technical Architecture

The Live Quiz Platform is built on a modern, well-established web application stack, selected for reliability, security, and the ability to run in cloud, on-premises, or hybrid environments without architectural rework.

17.1 Technology Stack

Layer	Technology	Notes
Frontend	React (single-page application)	Also deployable with Next.js where required by the client's environment
Backend	Node.js with Express.js	Exposes a REST API consumed by the frontend
Authentication	Token/session-based authentication	JWT or an equivalent secure token-based system
Database	PostgreSQL (recommended)	Enterprise database with row-level, per-organization data isolation; MongoDB may be considered depending on the final architecture agreed with [Client Name]
Real-time layer	WebSockets	Used for participant joining, live answers, and leaderboard updates; timing and scoring are server-authoritative
Hosting	Cloud, on-premises, or hybrid	See Section 19 for the full comparison of deployment options

17.2 Architecture Diagram

The diagram below shows the platform's core request and real-time data flow, followed by the three supported deployment variants.



Browser clients communicate with the Express.js API and WebSocket server over HTTPS and secure WebSockets (WSS); the application server is the sole component with direct database access.

Deployment Variant	Application Layer (Frontend + Express.js API)	Database Layer
Cloud / SaaS	Altanfith-managed cloud environment	Altanfith-managed cloud environment
On-Premises	[Client Infrastructure]	[Client Infrastructure]
Hybrid	Altanfith-managed cloud environment	[Client Infrastructure] (sensitive data remains client-side)

18. Real-Time Architecture

Live sessions depend on the platform's real-time layer behaving predictably under load, and behaving fairly for every participant regardless of network conditions. This section describes how that is achieved.

18.1 Session Flow

A live session follows a consistent, server-driven flow from join through to the final leaderboard.

Step	What happens
1. Join	Participants connect over a WebSocket session as they scan a QR code, follow a direct link, or enter a session PIN; the host's lobby view updates in real time as each participant joins
2. Question broadcast	When the host opens a question, the server broadcasts the question to all connected participants at the same moment
3. Answer submission	Participants submit answers over the WebSocket connection while the question is live; the correct answer is never sent to a participant's device before the question closes
4. Server-side scoring	The server, not the client, determines when the question closes, evaluates each submitted answer, and calculates the resulting score
5. Leaderboard push	Updated rankings are pushed to the host and participant devices in real time once scoring is complete

18.2 Server-Authoritative Timing, Reconnection, and Why It Matters at Scale

Question timing, closure, and scoring are controlled by the server, not by any individual device: participant and host devices render a countdown, an open or closed question, and an updated leaderboard, but they never decide when a question opens or closes and are never trusted with the correct answer before it is safe to reveal it. Where a participant's device loses connectivity mid-session — common in venues with variable Wi-Fi or mobile coverage — the platform automatically attempts to reconnect the session rather than dropping the participant. For a session of the scale described in Section 24, this combination is what keeps a large, live audience in a fair, synchronized session rather than a collection of individually drifting client experiences.

19. Deployment Options

[Client Name] can choose from three deployment options, so that the choice of hosting model reflects data residency, network, and cybersecurity requirements rather than a one-size-fits-all default.

	Option 1 — SaaS / Cloud	Option 2 — On-Premises	Option 3 — Hybrid
Hosted by	Altanfith	Client infrastructure	Split between Altanfith and client infrastructure
Includes	Cloud hosting, server & database management, backups, security updates, monitoring, maintenance, technical support, version upgrades, AI usage allowance	Application deployment, backend installation, database configuration, environment setup, initial cybersecurity configuration, documentation, training, knowledge transfer	Application components hosted by Altanfith; sensitive data/database remain in the client environment
Commercials	Annual subscription; lower initial investment; fastest implementation	Project fee; optional source-code handover; optional annual maintenance contract	Per final design
Client provides	Browser access only	Hardware, OS, internal networking, data-center requirements (unless in scope)	Per final design

19.1 Cloud / SaaS

Altanfith hosts and manages the full application and database layer, so [Client Name] requires only browser access. Hosting, backups, monitoring, security updates, and version upgrades are managed by Altanfith as part of the subscription (Section 33, Model A).

19.2 On-Premises

The Express.js backend runs on the client's own internal servers or private cloud — a public cloud environment is not required. The platform's backend has been built to run within a range of internal environments, including:

Supported Environment	
Client data center	Internal servers
Private cloud	Virtual machines

Docker containers	Kubernetes
Linux servers	Windows Server (subject to technical validation)
Hybrid combinations of the above infrastructure types	

On-premises deployment is **subject to infrastructure assessment**. Network policies, database selection, internal DNS, SSL certificates, firewall configuration, backup procedures, and **[Client Name]**'s cybersecurity requirements are all confirmed during this assessment before deployment proceeds. Altanfith does not promise universal compatibility with every client environment in advance of this assessment.

19.3 Hybrid

A hybrid deployment splits the platform between Altanfith-hosted application components and client-hosted, sensitive data storage. The final design of a hybrid deployment depends on **[Client Name]**'s specific security and integration requirements, and is agreed during discovery and infrastructure assessment rather than fixed in advance.

20. Cybersecurity

The Live Quiz Platform is designed with cybersecurity as a foundational concern rather than an add-on, and its controls are developed with awareness of the OWASP Top 10. At the same time, we are deliberate about how we describe our security posture: the statements in this section describe how the platform is designed and configured, not a formal certification, and final acceptance of the security architecture remains subject to **[Client Name]**'s own cybersecurity assessment.

Domain	Controls
1. Encryption & transport	HTTPS/TLS; encryption in transit; encryption at rest where applicable; backup encryption
2. Identity & access	Secure (bcrypt) password hashing; role-based access control (RBAC); least-privilege access; strong password policy; multi-factor authentication (Optional); session management; token expiration; secure API authentication; user account deactivation; regular access reviews; role-validated host-only operations
3. Application security	API rate limiting; input validation; output encoding; protection against common web vulnerabilities; secure development practices; developed with awareness of the OWASP Top 10; secure API configuration
4. Audit & monitoring	Audit logs of administrative actions; login logs; failed-login monitoring; data-access logging; monitoring & alerting; log retention ([Data Retention Period])
5. Data lifecycle & resilience	Backup policy; backup retention; disaster recovery planning; business continuity planning; data retention rules; data deletion and export procedures — expanded in Section 21
6. Infrastructure & operations	Secure environment configuration; firewall configuration; network segmentation (on-premises); per-tenant, row-level data isolation in PostgreSQL; database access restrictions; secrets management; environment-variable management; secure source-code repository; separated dev/test/prod environments
7. Process	Change management; incident response; security incident notification; data-breach escalation; secure release process; security patching; vulnerability assessment; penetration testing (Optional service); client-approved security architecture

These controls are designed to align with recognized good practice and with awareness of the OWASP Top 10 — not represented as certified or compliant. Final acceptance is confirmed through **[Client Name]**'s own security assessment (Phase 3, Section 25); the security architecture is only finalized once client-approved. Formal penetration testing is available as an **optional service**, quoted separately (Section 33); it is not included in the base engagement.

21. Data Protection

Data protection builds on the data lifecycle and resilience controls introduced in Section 20 (Domain 5), and describes, in more detail, how **[Client Name]**'s data is backed up, retained, deleted, and exported. No guarantee of zero downtime or zero data loss is made; the platform is designed to align with sound backup, retention, and recovery practice within the parameters agreed with **[Client Name]**.

Aspect	Approach
Backups	Operated under a defined backup policy; frequency, encryption, and retention arrangements are confirmed as part of the applicable deployment option (Section 19) and the commercial agreement
Retention	Session results, audit logs, and participant records are retained for [Data Retention Period] , configurable to support [Client Name] 's internal data governance or regulatory requirements, subject to confirmation during discovery
Deletion	Follows a defined procedure: soft-delete with archive is available at the content level (Section 13.5); permanent deletion procedures, including at contract termination, are defined in the commercial agreement
Export	Available on request, and provided as a standard step at contract termination, in an agreed format, so [Client Name] is not dependent on continued platform access to retain its own data
Disaster recovery & business continuity	Addressed as part of the deployment design and configured to support [Client Name] 's requirements; specific recovery time/point objectives are confirmed during discovery and infrastructure assessment and depend on the deployment option selected

22. Data Ownership

Ownership of data and intellectual property is set out clearly, so that **[Client Name]** understands exactly what it owns, what Altanfith owns, and how this is affected by the commercial model selected.

- Client data remains the property of **[Client Name]** at all times.
- Altanfith Aldwaliah SPC claims no ownership of client operational data.
- Data export is provided at contract termination, in an agreed format.
- Retention and deletion terms are defined in the commercial agreement.
- For on-premises deployments, data may remain entirely within **[Client Infrastructure]**.
- Intellectual property ownership depends on the commercial model selected (see Section 33).
- Standard platform IP remains with Altanfith Aldwaliah SPC unless transferred by separate agreement.
- Client-specific content — including questions, participant data, and reports — remains the property of **[Client Name]**.
- Source-code ownership is not automatically included, and applies only where it is expressly stated in the commercial agreement.

23. Integration Options

The Live Quiz Platform exposes REST APIs that support integration into **[Client Name]**'s existing systems. Integrations are delivered as scoped work under the Enterprise Custom Development Partnership commercial model (Model C, see Section 33), rather than assumed as part of the base platform, and their final scope is always subject to the availability and technical constraints of the relevant third-party API.

Integration	Description	Status
Single sign-on (Microsoft Entra ID / Active Directory)	Authentication against the client's existing identity provider, with roles mapped to directory groups	Optional
HR systems	Integration with the client's HR system, for example for user or organizational-unit synchronization	Optional
Learning Management System (LMS)	Integration with the client's LMS for training-related workflows	Optional
Microsoft Teams	Integration supporting session distribution or notification via Microsoft Teams	Optional
Power BI	Integration exposing platform data for consumption in Power BI dashboards	Optional
[Integration System]	Any additional client system identified during discovery	Optional

All integrations listed above are optional and subject to the availability, documentation, and technical constraints of the relevant third-party API; final integration scope is confirmed during discovery (see Section 25, Phase 1) and delivered under Phase 8 (Integration Development) of the implementation methodology.

24. Performance and Scalability

Performance and scalability claims for the Live Quiz Platform are grounded in what has actually been observed, rather than presented as an unqualified guarantee.

The platform has supported **more than 100 participants in a single live session in real production use**, as described in Section 12. In addition to this production track record, internal load exercises have separately validated **400 concurrent participant joins** against the production configuration, giving **[Client Name]** a tested reference point beyond day-to-day operational use.

Beyond these two reference points, the following considerations apply to any deployment for **[Client Name]**:

- Final concurrency targets are confirmed during discovery, based on **[Concurrent Participants]**.
- Load testing is recommended ahead of any enterprise deployment with a large expected audience.
- Cloud scaling and infrastructure sizing are based on the expected usage profile agreed with **[Client Name]**.
- For on-premises deployments, real-time performance depends on the client's own infrastructure and network configuration.
- Real-time functionality requires proper network and server configuration to perform as designed.
- Performance monitoring may be included as part of the applicable support and maintenance package (see Section 31).

Altanfith Aldwaliah SPC does not make claims of unlimited scalability. Performance at higher concurrency levels than those referenced above should be validated through load testing scoped to **[Client Name]**'s specific requirements before a large-scale go-live.

PART FOUR

How Altanfith plans, governs, and delivers the engagement

25. Implementation Methodology

Altanfith Aldwaliah SPC delivers the Live Quiz Platform through a structured, 17-phase methodology, designed to take **[Client Name]** from initial requirements through to a supported, live production system with clear exit criteria at each stage.

Phase	Purpose & Key Activities	Exit Criteria
1. Discovery & Requirements Gathering	Structured workshops and stakeholder interviews establish functional, branding, and organizational requirements	Documented and agreed requirements
2. Technical Assessment	Reviews [Client Infrastructure] , network, and hosting environment	Assessment report; confirmed deployment option
3. Cybersecurity Review	Engages the client's security team on the proposed architecture and controls (Section 20)	Client-approved security architecture
4. Solution Design	Translates requirements into a detailed solution design, incl. configuration, branding, and custom/integration scope	Approved solution design document
5. UI & Branding Customization	Applies the client's logo, color palette, themes, and UI text	Branded environment reviewed by client
6. Configuration	Configures organizations, departments, user roles, and platform settings	Configured environment ready for build/test
7. Custom Development	Builds agreed custom features, workflows, or reports	Custom development complete and reviewed
8. Integration Development	Builds and configures agreed integrations (Section 23)	Integrations functioning against client systems
9. Internal Testing	Functional and regression testing across the configured environment	Test results reviewed; defects triaged
10. Security Testing	Security-focused testing against the Section 20 controls	Testing complete; remediation applied

11. User Acceptance Testing	Client tests the platform against agreed acceptance criteria (Section 30)	UAT sign-off (Milestone M5)
12. Data Migration (if required)	Migrates agreed legacy data into the platform, where in scope	Migrated data validated against source
13. Deployment	Deploys the platform to the production environment	Production environment live and verified
14. Training	Administrator, quiz-creator, and support-team training (Section 29)	Training completed; materials handed over
15. Go-Live	Platform goes live for production use, following successful UAT	Go-live checklist completed (Milestone M6)
16. Hypercare	Heightened, close-support monitoring immediately following go-live	Hypercare completed; no unresolved critical issues
17. Transition to Support	Hands the platform from the project team to ongoing support (Section 31)	Support handover complete (Milestone M7)

26. Project Timeline

The table below presents a sample timeline of 6–12 weeks, mapped against the phases described in Section 25.

Weeks	Phases
W1–2	Discovery & Requirements Gathering; Technical Assessment; Cybersecurity Review
W2–4	Solution Design; UI & Branding Customization
W3–7	Configuration; Custom Development; Integration Development
W7–9	Internal Testing; Security Testing; User Acceptance Testing
W9–10	Deployment; Training
W10	Go-Live
W10–12	Hypercare; Transition to Support

26.1 Milestones

Milestone	Description	Indicative Completion	Sign-off by
M1	Requirements sign-off	W2	Client Project Manager
M2	Solution design approval	W4	Client Project Manager
M3	Configured & branded environment ready	W7	Altanfith Project Manager
M4	Integrations complete	W7	Altanfith Technical Lead
M5	UAT sign-off	W9	Client Project Manager

M6	Go-live	W10	Project Sponsor
M7	Hypercare exit / support handover	W12	Client Project Manager & Altanfith Project Manager

This timeline is indicative only. The actual project timeline for **[Client Name]** depends on the final agreed scope, the number and complexity of integrations, and the timeliness of client approvals at each milestone.

27. Project Governance

The engagement is governed through a clearly defined set of roles and a regular reporting cadence, so that progress, issues, and decisions are visible to both parties throughout delivery.

Role	Responsibility
Project Sponsor (client)	Owns the business case and provides executive-level decisions and escalation support
Client Project Manager	Coordinates client-side stakeholders, approvals, and resourcing
Altanfith Project Manager	Coordinates delivery, reporting, and issue/risk management on behalf of Altanfith
Technical Lead (Altanfith)	Owns technical design, development, and integration decisions
Cybersecurity Representative (client)	Reviews and approves the security architecture and controls
Business Owner (client)	Confirms functional requirements and acceptance criteria on behalf of the business

Governance is supported by a consistent set of cadence and mechanics: weekly status meetings; regular progress reporting; a maintained issue log and risk log; a formal change-request management process for scope changes; an escalation process running from project manager to project sponsors; and a documented sign-off process at each milestone described in Section 26.

28. Deliverables

The table below lists the deliverables produced over the course of the engagement.

Deliverable	Description	Included?
Requirements document	Documented, agreed functional and non-functional requirements	✓
Solution design document	Detailed design of the configured and/or customized solution	✓
Configured platform	The platform configured per the agreed solution design	✓
Customized UI	Branding and UI customization applied per Section 25, Phase 5	✓
Deployed frontend	The React frontend deployed to the agreed environment	✓
Deployed Express backend	The Express.js API and WebSocket server deployed to the agreed environment	✓
Database setup	Database configured and provisioned per the solution design	✓
Security configuration	Security controls configured per the client-approved security architecture	✓
User roles configured	Role-based access configured per the agreed role model	✓
Administrator account setup	Initial administrator accounts provisioned	✓
Reports	Reporting and analytics configured per Section 16	✓
Training material	Materials used in the training described in Section 29	✓

User manual	End-user reference documentation	✓
Administrator manual	Administrator reference documentation	✓
Deployment guide	Documentation of the deployed environment	✓
Backup guide	Documentation of backup procedures for the deployed environment	✓
Support handover	Formal handover to the ongoing support team (Phase 17)	✓
UAT plan	Agreed User Acceptance Testing plan and criteria (Section 30)	✓
Test report	Summary of internal, security, and acceptance testing outcomes	✓
Go-live checklist	Checklist confirming readiness for production go-live	✓
Source code	Provided only where included in the selected commercial model	Only if included in the commercial model
API documentation	Provided where included in scope	If included

29. Training and Knowledge Transfer

Training is provided so that **[Client Name]**'s own team can operate, administer, and support the platform confidently after go-live.

Training Item	Description	Status
Administrator training	Training for administrators on platform configuration and management	Included
Quiz-creator training	Training for staff authoring quizzes and question banks	Included
Support-team training	Training for the client's internal support or IT team	Included
User manuals	Reference documentation for end users	Included
Quick-start guide	Condensed getting-started reference	Included
Recorded training	Recorded training sessions for later reference or onboarding new staff	Optional
Train-the-trainer	Enables client-side trainers to deliver ongoing internal training	Optional
Arabic and English training material	Bilingual training material	Optional

30. Testing and Acceptance

The platform passes through three layers of testing before it is accepted for production go-live. Altanfith performs internal functional and regression testing (Phase 9) across the configured environment, followed by security-focused testing (Phase 10) against the controls described in Section 20, ahead of client security review and sign-off. User Acceptance Testing (Phase 11) is then carried out against a client-approved UAT plan and agreed acceptance criteria, so that **[Client Name]** — not Altanfith alone — confirms that the configured platform meets its requirements before go-live.

Any defects identified during internal, security, or acceptance testing are triaged using the same four-level severity model used for support (Section 32): Critical, High, Medium, and Low. UAT sign-off is Milestone M5 in the project timeline (Section 26), and it gates go-live: production rollout depends on successful UAT.

31. Support and Maintenance

Ongoing support is delivered under one of three packages, so that **[Client Name]** can select the level of coverage and responsiveness that matches its operational needs.

Package	Includes
Bronze	Oman business-hours support; email support; bug fixes; standard updates; best-effort response
Silver	Everything in Bronze, plus: priority support; faster response times; monthly system health review; performance optimization; minor improvements; remote assistance
Gold	Everything in Silver, plus: critical incident support; dedicated support contact; priority issue resolution; proactive monitoring; security monitoring; quarterly service review; emergency support; 24/7 support (Optional); dedicated development allocation (Optional)

32. SLA

Service levels are defined by severity, with response time and target resolution time tracked as two separate measures — response confirms that an issue has been received and is being actioned; resolution is the target time to resolve it, and is not a guarantee.

Severity	Definition	Response Time	Target Resolution Time
1 — Critical	Platform down / live session blocked for all users	[Response Time]	[TBC]
2 — High	Major function degraded, no workaround	[Response Time]	[TBC]
3 — Medium	Limited impact, workaround exists	[Response Time]	[TBC]
4 — Low	Cosmetic / question / minor request	[Response Time]	[TBC]

Response and target resolution times are separate measures: response time confirms an issue has been received and triaged; target resolution time is a goal, dependent on the agreed SLA, and is not a guaranteed outcome. Actual values depend on the support package selected (Section 31) and the SLA agreed with **[Client Name]**; 24/7 coverage applies only where the Gold 24/7 option has been purchased.